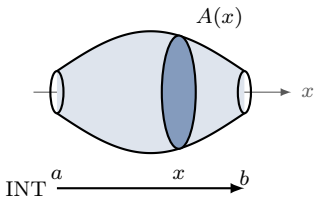
L4.2a Volumes by Slicing

Disks and cross-sections — what you need to know

VOLUME BY SLICING



$$V = \lim_{n\to\infty} \sum_{i=1}^{n} A(x_i^*)\,\Delta x = \int_a^b A(x)\,dx$$

$A(x)$ is the area of the cross-section at x, continuous on $[a, b]$

one slice: $V(S_i) \approx A(x_i^*)\,\Delta x$

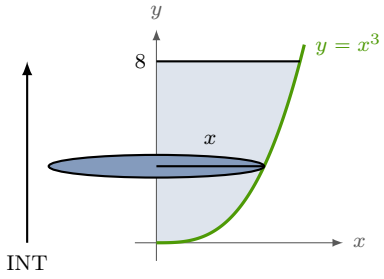
"the definition of volume: a sum through the slices, each one area times thickness"

Ex. Find the volume of a cylinder of radius r and height h. Every slice has $A(x) = \pi r^2$, so $V = \int_0^h \pi r^2\,dx = \boxed{\pi r^2 h}$.

Disks about the x-axis and about the y-axis

	About the x-axis	**About the y-axis**
slices perpendicular to	the x-axis, thickness Δx	**the y-axis, thickness Δy**
radius	$y = f(x)$	**x, written in terms of y**
area of a slice	$A(x) = \pi\,[f(x)]^2$	$A(y) = \pi x^2$
limits are	x-values	**y-values**
volume	$\int_a^b \pi\,[f(x)]^2\,dx$	$\int_c^d \pi x^2\,dy$

DISKS



$$\boxed{A = \pi\,(\text{radius})^2}$$

"the radius is the distance from the axis to the curve"

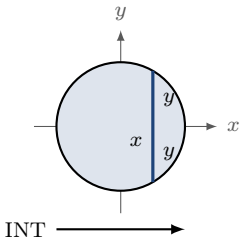
Ex. Rotate the region bounded by $y = x^3$, $y = 8$ and $x = 0$ about the y-axis. Find the volume.

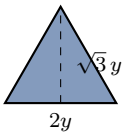
Slice perpendicular to the y-axis, so $V = \int_0^8 A(y)\,dy$. Invertible on the domain? Yes: $x = y^{1/3}$.

$$A(y) = \pi\left(y^{1/3}\right)^2 = \pi y^{2/3}$$

$$V = \pi \int_0^8 y^{2/3}\,dy = \tfrac{3\pi}{5}\,y^{5/3}\Big|_0^8 = \tfrac{3\pi}{5}\cdot 2^5 = \boxed{\tfrac{96\pi}{5}}$$

KNOWN CROSS-SECTIONS





$$V = \int_a^b A(x)\,dx$$

"the side of each slice is the width of the base at x"

square, side s
$A = s^2$

semicircle, diameter s
$A = \frac{1}{2}\pi\left(\frac{s}{2}\right)^2 = \frac{\pi}{8}s^2$

equilateral triangle, side s
$A = \frac{1}{2}s \cdot \frac{\sqrt{3}}{2}s = \frac{\sqrt{3}}{4}s^2$

Ex. A solid has a circular base of radius 1. Cross-sections perpendicular to the base are equilateral triangles. Find the volume.

Put the base on $x^2 + y^2 = 1$. At x the side is $2y$, with $y = \sqrt{1-x^2}$.

$$h = \sqrt{(2y)^2 - y^2} = \sqrt{3}\,y \qquad A(x) = \tfrac{1}{2}(2y)\left(\sqrt{3}\,y\right) = \sqrt{3}\,y^2 = \sqrt{3}\left(1 - x^2\right)$$

$$V = \int_{-1}^{1} \sqrt{3}\left(1 - x^2\right)dx = 2\sqrt{3}\int_0^1 1 - x^2\,dx = 2\sqrt{3}\left(1 - \tfrac{1}{3}\right) = \boxed{\tfrac{4\sqrt{3}}{3}}$$

WATCH OUT

✗ ~~$A(x) = (\pi f(x))^2$~~ → $A(x) = \pi\,[f(x)]^2$ — **only the radius is squared**

✗ ~~about the y-axis, $A(y) = \pi y^2$~~ → **solve the curve for x first — the radius is x**

✗ ~~limits in x for slices perpendicular to the y-axis~~ → **limits in y — you are integrating dy**

✗ ~~the side of the slice is y~~ → **the base runs from $-y$ to y — the side is $2y$**